



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **EEE G622**
Course Title : **Advanced Digital Communication**
Instructor-in-Charge : **Dr. Sharda Tripathi**
Tutorial/Practical Instructors: **Mr. Sudhir Kumar (RS)**

1. Course Description: Introduction to digital communication systems; signal characterization and representation of bandpass and low-pass signals; signal space analysis; fundamentals of source coding and entropy; digital modulation techniques and constellation design; optimum receivers for additive white Gaussian noise channels; maximum likelihood (ML) and maximum a posteriori (MAP) detection; symbol error probability analysis; non-coherent detection methods; carrier and symbol synchronization techniques; communication over band-limited channels; ISI, Nyquist criteria, and signal waveform design, equalization; multicarrier modulation systems, effects of intercarrier interference (ICI) and carrier frequency offset (CFO); channel capacity limits and fundamentals of channel coding.

2. Course Scope and Objective:

1. To provide fundamental knowledge to model, design, and analyze advanced digital communication systems, with applications in both industry and R&D.
2. To reinforce theoretical concepts through practical assignments using MATLAB, Python, and USRP.
3. Requires prior knowledge of Linear Systems, Probability Theory, Stochastic Processes, and basic Communication Systems.

3. Text Books:

[T1]: “Digital Communications”, by John G. Proakis, Masoud Salehi, Ed. 5, McGrawHill, 2008.

4. Reference Books:

[R1]: R. Gallager “Principles of Digital Communication”, Ed. 1, Cambridge University Press, 2008.

[R2]: Wozencraft & Jacobs, “Principles of communication engineering”, John Wiley, 1965.

[R3]: U. Madhow, “Fundamentals of Digital Communication”, Ed. 1, Cambridge University Press, 2008.

[R4]: “Communication Systems” by Simon Haykins, fourth edition, John Wiley, 2000.

[R5]: “Digital Communications: Fundamentals and Applications” by Bernard Sklar, Ed. 2, Pearson Education, 2002.



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5. Course Plan:

Module No.	Lecture Session	Reference	Learning Outcomes
(0): Introduction to Digital Communication	1-2: Principles of communication, elements of digital communication system, sources of information, communication channels and their characteristics, modulation process, history of digital communication	T1: chapter 1	Motivation to study and fundamental basis required for the course
(I): Deterministic and random signal analysis	3-5: Signal spaces, inner product and orthogonal expansion, Gram Schmitt orthogonalization, bandpass/lowpass signal representation	T1: chapter 2, chapter 3	Review of mathematical tools required to build the analytical understanding
(III): Digital modulation techniques	6-9: Coding of analog sources, Memoryless modulation – PAM, phase modulation, quadrature amplitude modulation, FSK, PPM, bi-orthogonal modulation, simplex signal set; Modulation with memory – CPFSK	T1: chapter 3	Study of digital modulation schemes and the power spectrum of digitally modulated signals
(IV) : Optimum receiver structures	10-13: Correlation demodulator, matched filter demodulator, optimum detector, ML detector, MAP detector.	T1: chapter 4	Design of optimum receivers for channels perturbed by AWGN.
(V): Performance analysis over AWGN channel	14-20: Error probabilities for band limited signaling, power limited signaling, optimal detection in presence of uncertainty – non coherent detection	T1: chapter 4	Study the error rate performance of optimum receivers in AWGN
(VI): Carrier and symbol synchronization	21-23: Signal parameter estimation, carrier phase estimation, symbol timing estimation – early late gate synchronizers	T1: chapter 5	Carrier phase estimation and time synchronization based on maximum likelihood criterion
(VII) : Digital communication through bandlimited channels	24-30: Signal design for band limited channels, probability of error for zero ISI and system design in the presence of channel distortion, equalizer design	T1: chapter 9, chapter 10	Characterization and system design for bandlimited channels
(VIII): Source Coding	31-33: Fundamentals of source coding, entropy, and data compression.	T1: chapter 6	To apply the principles of source coding, entropy, and data compression for efficient information representation and transmission.



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(IX): Channel capacity and coding	34-39: Channel models and capacity, error control coding – linear block codes, convolutional codes	T1: chapter 7, chapter 8	Introduction to topics in information theory and their application in digital communication system
(X): Wireless communication	40- 42: Multicarrier systems, fundamentals of OFDM and spread spectrum signals, effects of ICI and carrier frequency offset (CFO).	T1: chapter 11, chapter 12	Understanding multicarrier systems, their modulation and demodulation methods

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 mins.	25	05-10-2026 FN2	Closed Book
Comprehensive Examination	3 hrs	35	11-12-2026 AN	Closed/Open Book
Quizzes	15 mins	10	Spread Across Semester	Closed Book
Regular lab assignments	-	15	Spread Across Semester	Open Book
Research Seminars / Research Project	-	15	Spread Across Semester	Open Book

*Open Book: Note that Open Book in the comprehensive exam is only limited to accessing the textbook (as mentioned in the handout), and handwritten class notes.

7. Chamber Consultation Hour: Will be announced in class

8. Notices: All notices will be displayed on the Learning Management System (LMS-NALANDA)

9. Make-up Policy: No make-up will be given for Quizzes and Lab evaluatives. However, for other components, make-up will be allowed ONLY in genuine cases. In such cases students must produce enough proof or must have taken the prior permission from the IC.

10. Note (if any):

- All tests and the exams make use of only a non-programmable calculator.
- Strict action, including awarding of zero marks, will be taken in case of plagiarism of the lab codes and/or lab reports. All submissions will be checked by a plagiarism detection tool.

Instructor-in-charge
Course No. EEE G622